



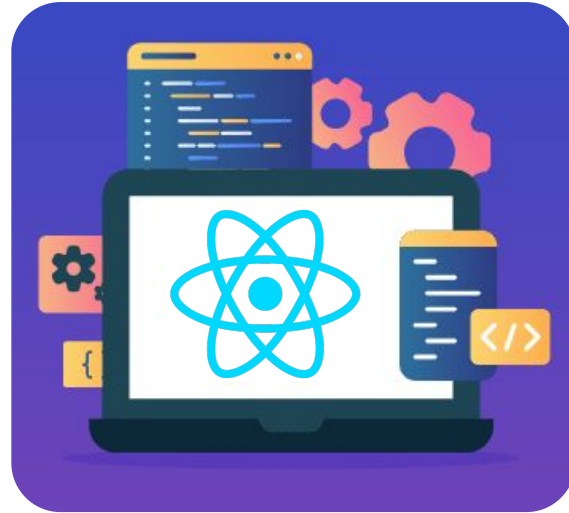
WELCOME





Introduction to React JS

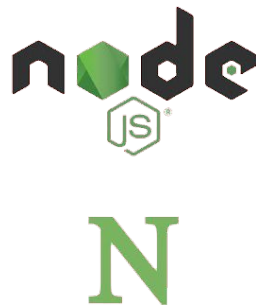
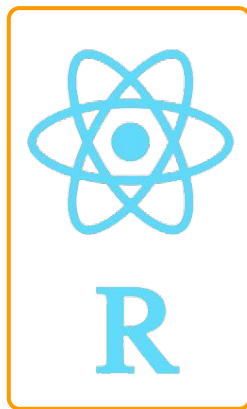
React JS



MERN Stack

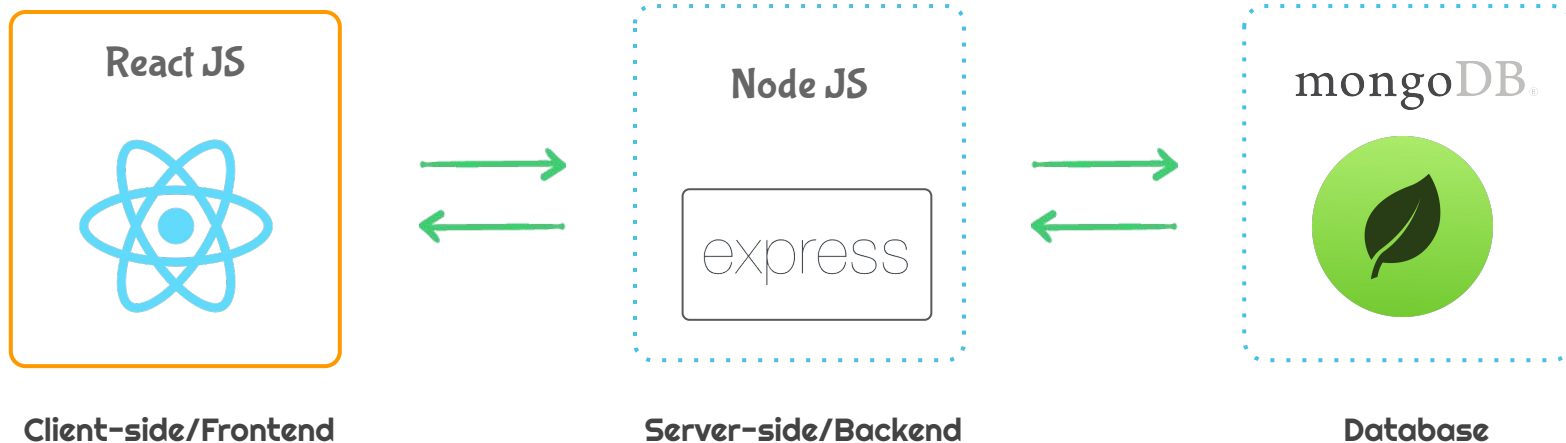
React JS

MERN stands for MongoDB, Express JS, React JS and Node JS



MERN Stack

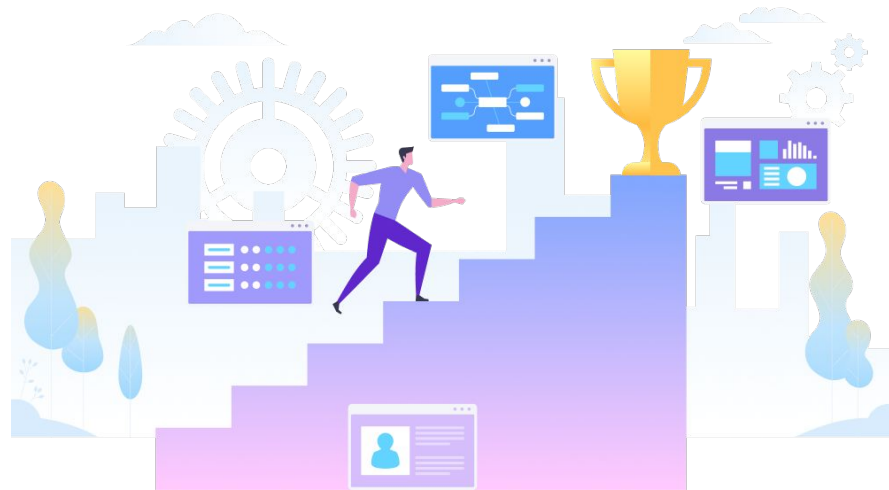
Application Flow



Topics to be Covered

React JS

- ✓ Introduction to React & JSX
- ✓ Components and Props
- ✓ Lists and Keys
- ✓ State and Events
- ✓ Handling Side Effects
- ✓ Routing using react-router
- ✓ Authentication & Context



Agenda for Today's Session

Introduction to React JS

Introduction to React JS



MERN Stack

React JS

React.createElement()
ReactDOMClient.createRoot()

Introduction

React JS

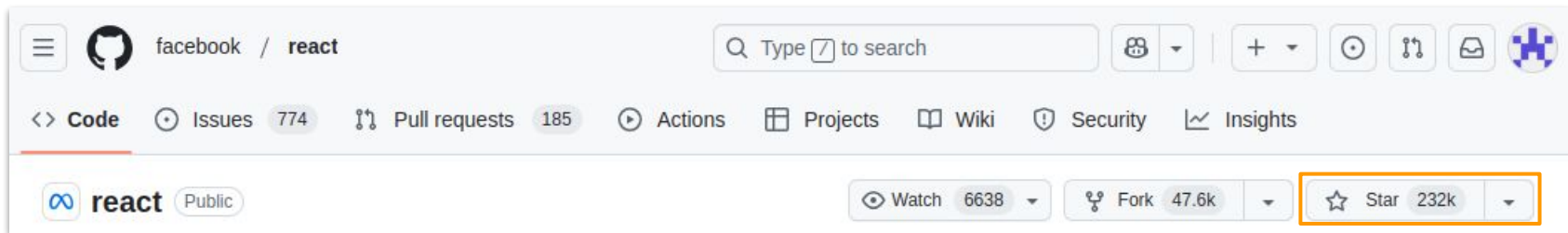
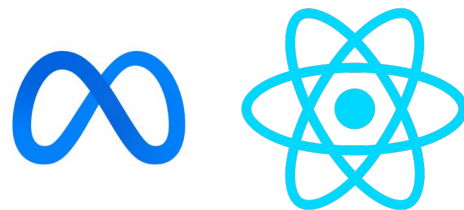
React JS is an **open-source** JavaScript library used to build **user interfaces**



Introduction

React JS

React JS was developed by Facebook (Meta)



<https://github.com/facebook/react>

Why React JS?

- ✓ Performant websites
- ✓ Less lines of code
- ✓ Improves readability of code
- ✓ Less time consuming
- ✓ Open Source
- ✓ Reusable code



Top Companies using React JS



NETFLIX



udemy

LaunchDarkly →

instacart



slack



many more...

Source: [Stackshare.io](https://stackshare.io)

Introduction

Advantages of React JS



Easy to Learn



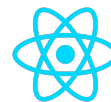
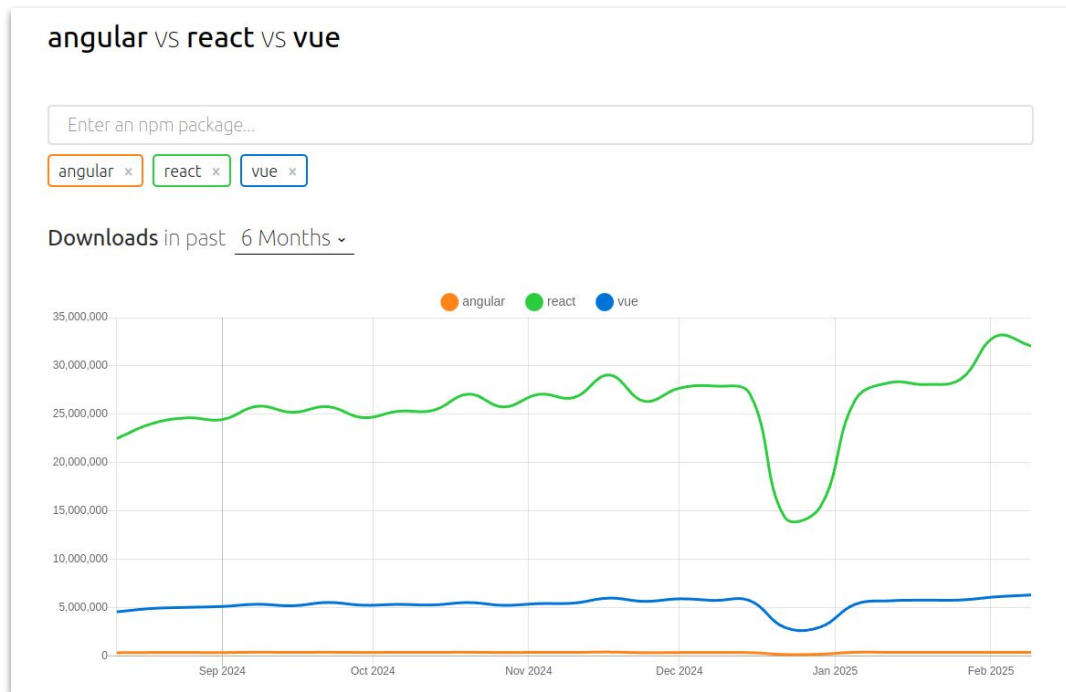
Large Community



Developer Toolset

React JS

Download Trends - React vs Others



Source: npmtrends.com

Session Initial Setup



Code Playground Instructor 

< CONTENTS -|

Introduction to React JS and States

○ Your Journey With React

○ Setting up GitHub Co-pilot

● Introduction to React JS

○ Introduction to React JS | Reading Material

○ Components and Props

○ Components and Props | Reading Material

○ Lists & Keys

○ Lists and Keys | Reading Material

○ State & Events

○ State and Events | Reading Material

○ Common Mistakes

WELCOME

< 1 > ⋮

Google Slides

SUBMIT FEEDBACK

✓ MARKED AS COMPLETE

NEXT SECTION >

Notes

Code Playground

Instructions

Complete the below steps to create your development environment and start working on some amazing projects!

Step - 1 : Setting up IDE (one time process)

Follow these instructions [Click Here](#)

🕒 If you have already finished step-1 before, skip this and continue directly to step-2

Step - 2 : Setting up a Workspace for your project

Run the below command in your IDE to setup a workspace for the project discussed in the session

ccbp start RJSIVTXLho

COPY

Help



Introduction to React JS

Session Initial Setup



Search for 'Arrays'

HomeCode Playground

<CONTENTS

Introduction to React JS
9 hours, 30 mins

Introduction to React JS
50 mins

Introduction to React JS | Cheat Sheet
15 mins

Practice 1
30 mins

Submit Feedback

Ask doubt

NotesCode PlaygroundDiscussionsSlides

Instructions

Go through the below steps to complete this question:

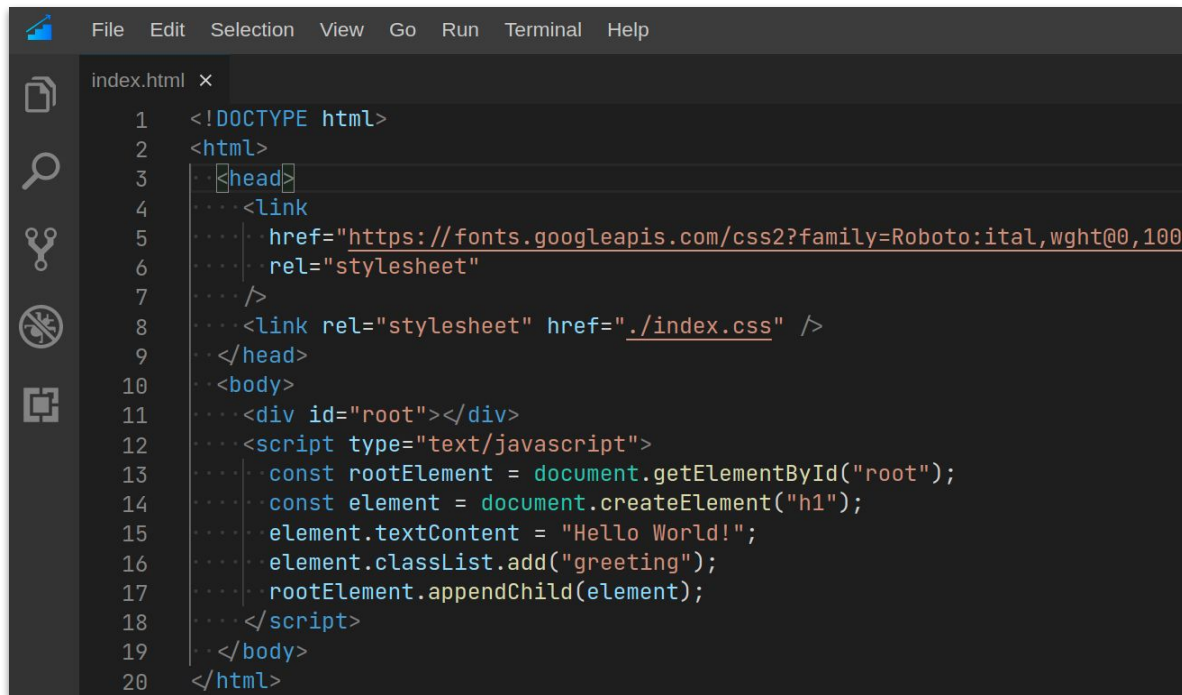
- Open IDE (It may take **1-2 minutes** to initialize the environment)
- Practice the concepts and skills you have learned during the session using IDE

OPEN IDE



Session Initial Code

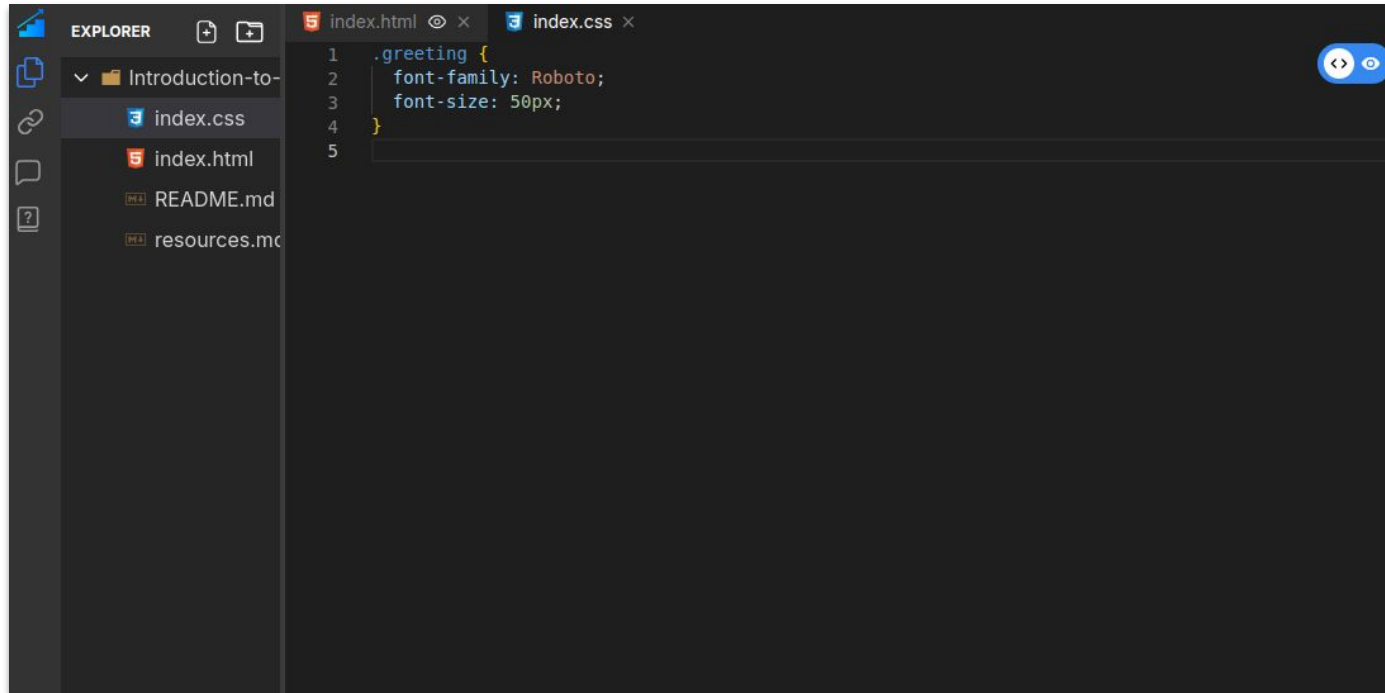
index.html



```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <link
5       href="https://fonts.googleapis.com/css2?family=Roboto:ital,wght@0,100
6       rel="stylesheet"
7     />
8     <link rel="stylesheet" href="._/index.css" />
9   </head>
10  <body>
11    <div id="root"></div>
12    <script type="text/javascript">
13      const rootElement = document.getElementById("root");
14      const element = document.createElement("h1");
15      element.textContent = "Hello World!";
16      element.classList.add("greeting");
17      rootElement.appendChild(element);
18    </script>
19  </body>
20 </html>
```


Session Initial Code

index.css



The screenshot shows a code editor interface with a dark theme. On the left, the 'EXPLORER' sidebar displays a file tree for a project named 'Introduction-to-'. The files listed are 'index.css', 'index.html', 'README.md', and 'resources.md'. The 'index.css' file is selected. The main editor area shows the content of 'index.css' with the following code:

```
1 .greeting {  
2   font-family: Roboto;  
3   font-size: 50px;  
4 }  
5
```

Line numbers 1 through 5 are visible on the left side of the code block. The editor tabs at the top show 'index.html' and 'index.css'.

Running JS in HTML

HTML

```
<!DOCTYPE html>
<html>
  ...
  <body>
    <div id="root"></div>
    <script type="text/javascript">
      const rootElement = document.getElementById("root");
      const element = document.createElement("h1");
      element.textContent = "Hello World!";
      element.classList.add("greeting");
      rootElement.appendChild(element);
    </script>
  </body>
</html>
```

`<script>` tag is used to include
JavaScript in HTML

`type` attribute specifies the
type of the script

Running JS in HTML

Checking Output

Open Terminal, run the below command,
and enter **y** when asked for confirmation

Terminal

```
root@123:/ ... /intro-to-react# npx serve
```

Running JS in HTML

Checking the Output

```
index.html
2 <html>
3 <head>
4 </head>
5 <body>
6 <div id="root"></div>
7 <script type="text/javascript">
8   const rootElement = document.getElementById("root");
9   const element = document.createElement("div");
10   element.textContent = "Hello World!";
11   element.classList.add("greeting");
12   rootElement.appendChild(element);
13 </script>
14 </body>
15 </html>
```

```
root@e4d10e2a0959:/home/nxtwave/introduction-to-react-js# npx serve

Serving!
- Local:    http://localhost:3000
- Network:  http://172.18.0.2:3000
```

Click on the link to Open the Server URL
in a New Tab to check your output



Regular JS

Output

Hello World!

Let's create
a Heading Element
using React JS

Creating a Heading Element using React JS

Adding React

HTML

```
<head>
  ...
</head>
<body>
  ...
  <script type="text/javascript" >
    import React from "https://esm.sh/react@19.1.1/?dev";
    import ReactDOMClient from "https://esm.sh/react-dom@19.1.1/client?dev";
  </script>
</script>
</body>
```

Creating a Heading Element using React JS

Adding React

HTML

```
<head>
  ...
</head>
<body>
  ...
  <script type="text/javascript" >
    import React from "https://esm.sh/react@19.1.1/?dev";
    import ReactDOMClient from "https://esm.sh/react-dom@19.1.1/client?dev";
  </script>
</script>
</body>
```

import is a JavaScript keyword used to bring in code from another file or external library

JavaScript Imports and Exports

import

- **import** is used when you want to bring in code from another file

export

- **export** is used when you want to share variables, functions or classes from one file so that other files can use them

** You will learn more about imports and exports in further classes*

Creating a Heading Element using React JS

Adding React

HTML

```
<head>
  ...
</head>
<body>
  ...
  <script type="text/javascript" >
    import React from "https://esm.sh/react@19.1.1/?dev";
    import ReactDOMClient from "https://esm.sh/react-dom@19.1.1/client?dev";
  </script>
</script>
</body>
```

This line imports the React library, which is hosted on the esm.sh CDN.

Creating a Heading Element using React JS

Changing Type Attribute

HTML

```
<head>
  ...
</head>
<body>
  ...
  <script type="module" >
    import React from "https://esm.sh/react@19.1.1/?dev";
    import ReactDOMClient from "https://esm.sh/react-dom@19.1.1/client?dev";
  </script>
</script>
</body>
```

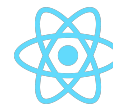
Creating a Heading Element using React JS

React.createElement()



JS

```
document.createElement("h1");
```



JS

```
React.createElement("h1");
```

Syntax

React.createElement(^{tag name}type, ^{properties}props, ^{Child nodes}children);

- ✓ **type** - div, h1, p etc
- ✓ **props** - className, onClick, id etc
- ✓ **children** - Represents the child nodes

(strings, numbers, nested react elements, etc)

Creating a Heading Element using React JS

React.createElement()

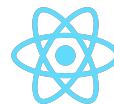
JS

```
const element = document.createElement("h1");  
element.textContent = "Hello World!";  
element.classList.add("greeting");
```



JS

```
React.createElement("h1", { className: "greeting"}, "Hello World!");
```



Creating a Heading Element using React JS

React.createElement()

HTML

```
<body>
  ...
  <script type="module">...
    import ReactDOMClient from "https://esm.sh/react-dom@19.1.1/client?dev";
    const elementProps = { className: "greeting" };
    const childrenContent = "Hello World!";
    const elementType = "h1";
    const element = React.createElement(elementType, elementProps, childrenContent);

  </script>
</body>
```

Creating a Heading Element using React JS

Displaying Element

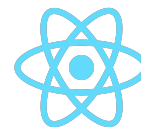
JS

```
const rootElement = document.getElementById("root");  
const element = document.createElement("h1");  
rootElement.appendChild(element);
```



JS

```
const rootElement = document.getElementById("root");  
const element = React.createElement(type, props, children);  
const root = ReactDOMClient.createRoot(rootElement);  
root.render(element);
```



Creating a Heading Element using React JS

Creating a Root Element

The `ReactDOMClient.createRoot()` lets us to create a root element to display React Elements inside a browser DOM

```
ReactDOMClient.createRoot(container);
```



Where to render

Creating a Heading Element using React JS

Displaying Element

The `root.render()` displays a React node into the React root's browser DOM node

What to render

```
root.render(reactElement);
```

Creating a Heading Element using React JS

Output

Hello World!

Creating a Heading Element using React JS

Displaying an Element

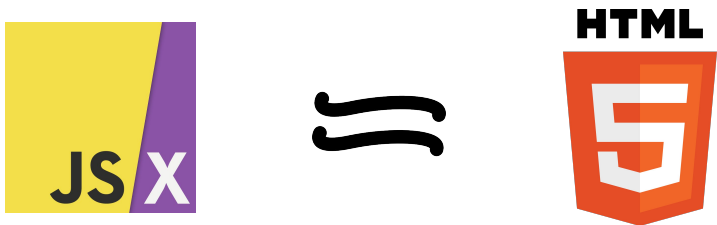
HTML

```
<body>
  <div id="root"></div>
  <script type="module">
    const elementProps = { className: "greeting"};
    const elementType = "h1";...
    const element = React.createElement(elementType, elementProps, childrenContent);
    const root = ReactDOMClient.createRoot(document.getElementById("root"));
    root.render(element);
  </script>
</body>
```

Create a Heading Element using React JS

JSX

React JS introduced a new **HTML** like syntax named as **JSX** to create elements



JSX

HTML like syntax

JSX

```
const element = <h1 className="greeting">Hello World!</h1>;
```

JSX

HTML like syntax

JSX

```
const element = <h1 className="greeting">Hello World!</h1>;
```

Compiles to

JS

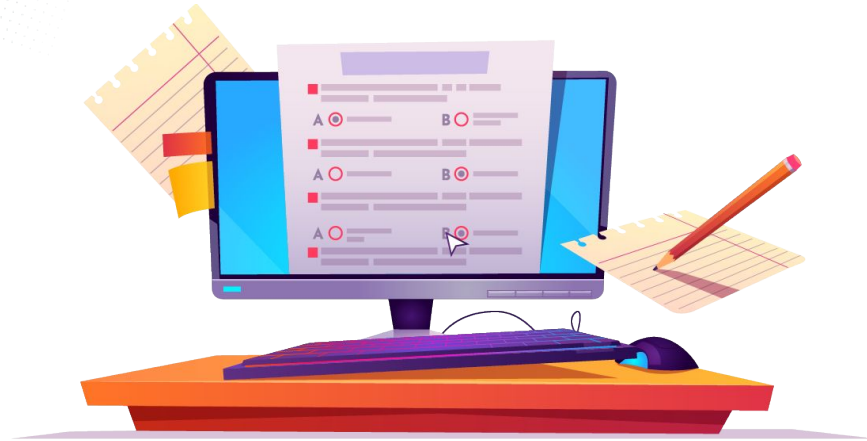
```
const elementProps = { className: "greeting"};  
const childrenContent = "Hello World!";  
const element = React.createElement(elementType, elementProps, childrenContent);
```

Key Takeaways

- ✓ MERN Stack
 - Introduction to React JS
- ✓ React
 - `React.createElement()`
 - `ReactDOMClient.createRoot()`
 - `root.render()`



Practice



Next Session

Components & Props





THANK YOU





ALL THE BEST

